reflecting telescope (reflector) A telescope that uses a concave mirror to collect light, reflect light rays to a focus, and form an image of a distant object.

reflection nebula see *nebula*.

refracting telescope (refractor) A telescope that uses a lens to refract (bend) light rays in order to bring them to a focus and form an image of a distant object.

regolith A layer of loose rock, rocky fragments, and dust that covers the surface of a planet or planetary satellite.

regular cluster see *galaxy cluster*.

relativity Theories developed in the early part of the 20th century by Albert Einstein to describe the nature of space and time and the motion of matter and light. The **special theory of relativity** describes how the relative motion of observers affects their measurements of mass, length, and time. One of its consequences is that mass and energy are equivalent. The **general theory of relativity** treats gravity as a distortion of space-time associated with the presence of matter or energy. One of its consequences is that massive bodies deflect rays of light. See also *gravitational lensing*, *space-time*.

resonance A gravitational interaction between two orbiting bodies that occurs when the orbital period of one is an exact, or nearly exact, simple fraction of the orbital period of the other. For example, Jupiter's moon Io is in a 1:2 resonance with another of Jupiter's moons, Europa (Io's period is half of Europa's period). When a small object is in resonance with a more massive one, it experiences a periodic gravitational tug each time one of the bodies overtakes the other, the cumulative effect of which gradually changes its orbit.

retrograde motion **(1)** The apparent backward motion of a planet, from east to west relative to the background stars. For most of the time, a planet such as Mars or Jupiter will move from west to east relative to the stars (**direct motion**), but it will appear to reverse direction each time it is being overtaken by Earth (around the time of opposition). See also *opposition*. **(2)** Orbital motion in the opposite direction of that of Earth and the other planets of the solar system. **(3)** The motion of a satellite along its orbit in the opposite direction to that in which its parent planet is rotating.

retrograde rotation The rotation of a body around its axis in the opposite direction to the rotational motion of Earth, the Sun, and the majority of the planets. Viewed from above its North Pole, Earth rotates around its axis and revolves around the Sun, in a counterclockwise direction (**direct rotation**), whereas a planet with retrograde rotation spins in the opposite (clockwise) direction. The planets Venus, Uranus, and Pluto exhibit retrograde rotation.

rich cluster see *galaxy cluster*.

right ascension (RA) The angular distance, measured eastward, between the **first point of Aries** (where the Sun's path around the sky crosses the celestial equator from south to north) and a celestial body. It is expressed in hours, minutes, and seconds of time, where 1 hour is equivalent to an angle of 15°. Together with declination, it specifies the position of a body on the celestial sphere. See also *celestial sphere*, *declination*, *ecliptic, equinox*.

ring A flat distribution of small particles and lumps of material that revolves around a planet, usually in the plane of its equator. A **ring system** consists of a number of concentric rings surrounding a planet. The planets Jupiter, Saturn, Uranus, and Neptune each have a ring system.

rocky planet A planet (also called a terrestrial planet) that is composed mainly of rocks and has basic characteristics similar to Earth. Within the solar system, there are four rocky planets: Mercury, Venus, Earth, and Mars. See also *gas planet*.

rupes Scarps or cliffs on the surface of a planet or a satellite. See also *moon*.

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satellite A body that revolves around a planet, otherwise known as a "moon." An **artificial satellite** is an object deliberately placed in orbit around Earth or around another solar system body.

Schmidt–Cassegrain telescope A type of catadioptric telescope. Light enters the telescope tube through a thin corrector lens and is reflected from a concave mirror at the bottom of the tube toward a small convex mirror fixed to the inner face of the correcting lens. It is then reflected back down the tube, through a hole in the concave mirror, to a focus. This is a popular, compact design for small and moderate-sized telescopes. See also *catadioptric telescope*.

Schwarzschild radius see *black hole*.

semimajor axis see *ellipse*.

Seyfert galaxy A spiral galaxy with an unusually bright, compact nucleus that in many cases exhibits brightness fluctuations. First identified by American astronomer Carl Seyfert in 1943, Seyfert galaxies comprise one of the several categories of active galaxies. See also *active galaxy*.

shepherd moon A small natural satellite that, through its gravitational influence, confines orbiting particles into a well-defined ring around a planet. A pair of shepherd moons, where one is slightly closer to the planet than the other, can squeeze particles into particularly narrow rings.

sidereal orbital period see *orbital period*.

sidereal time A time system based on the apparent rotation of the celestial sphere. **Local sidereal time** is defined to be 0 hours at the instant the first point of Aries crosses an observer's meridian. The **sidereal day** corresponds to Earth's axial rotation period measured relative to the background stars, and is equal to 23 hours 56 minutes 4 seconds of mean (civil) time. See also *equinox, right ascension*.

singularity A point of infinite density into which matter has been compressed by gravity, and a point at which the known laws of physics break down. Theory implies that a singularity exists at the center of a black hole. See also *black hole*.

solar cycle A cyclic variation in solar activity (for example, the production of sunspots and flares), which reaches a maximum at intervals of about 11 years. Because the polarity pattern of magnetic regions on the Sun reverses every 11 years or so, the overall duration of the cycle is 22 years. The sunspot cycle is the 11-year variation in the number (and overall area) of sunspots. See also *solar flare*, *sunspot*.

solar eclipse see *eclipse*.

solar flare A violent release of huge amounts of energy—in the form of electromagnetic radiation, subatomic particles, and shock waves—from a site located just above the surface of the Sun.

solar mass A unit of mass equal to the mass of the Sun, which provides a convenient standard for comparing the masses of stars. One solar mass is equivalent to 1.96×10^{27} tons (1.989×10^{30} kg). Stellar masses range from about 0.08 solar masses up to about 100 solar masses.

solar nebula The cloud of gas and dust from which the Sun and planets formed. As the cloud collapsed, most of its mass accumulated at the center to form the Sun, whereas the rest flattened out into a disk within which planets were assembled by the process of accretion. See also *accretion*, *protoplanetary disk*.

solar system The Sun together with everything that revolves around it (the planets and their satellites, asteroids, comets, meteoroids, gas, and dust).

solar wind A stream of fast-moving, charged particles (predominantly electrons and protons) that escapes from the Sun and flows outward through the solar system like a wind.

solstice One of the two points on the ecliptic at which the Sun is at its maximum declination north or south of the celestial equator. On or around June 21 each year, the Sun reaches its greatest northerly declination. This is the Northern Hemisphere **summer solstice** (the **winter solstice** in the Southern Hemisphere). On or around December 22 each year, the Sun reaches its greatest southerly declination. This is the Northern Hemisphere winter solstice (the summer solstice in the Southern Hemisphere). See also *celestial equator*, *declination*, *ecliptic*.

space-time The four-dimensional combination of the three dimensions of space (length, breadth, and height) and the dimension of time. The concept that time and space are intimately linked, rather than (as Newton had believed) being separate entities, was proposed in 1908 by Hermann Minkowski and was incorporated into Albert Einstein's theories of relativity. See also *relativity*.

special theory of relativity see *relativity*.

spectral line A feature that appears at a particular wavelength in a spectrum. An **emission line** is a bright feature corresponding to the emission of light at that wavelength, whereas an **absorption line** is a dark feature corresponding to the absorption of light at that wavelength. See also *spectrum*.

spectral type A class into which a star is placed according to the lines that appear in its spectrum. The principal spectral types, arranged in decreasing order of temperature, are labeled O, B, A, F, G, K, M and are subdivided into numbers from 0 to 9. For example, the spectral type of the Sun is G2. See also *luminosity*, *spectral line*, *spectrum*.

spectroscopic binary see *binary star*.

spectroscopy The science of obtaining and studying the spectra of objects. Because the detailed appearance of a spectrum is influenced by factors such as chemical composition, density, temperature, rotation, velocity, turbulence, and magnetic fields, spectroscopy can reveal a wealth of information about the physical and chemical properties of, and processes occurring in, planets, stars, gas clouds, galaxies, and other kinds of celestial bodies. See also *spectrum*.

spectrum A beam of electromagnetic radiation spread out into its constituent wavelengths. A **continuous spectrum** is the unbroken spread of wavelengths emitted by a hot solid or liquid or a dense gas (the continuous spectrum of sunlight appears to human eyes as a rainbow band of colors). A hot, low-density gas emits light at particular wavelengths only; the resulting spectrum consists of bright **emission lines**, each of which corresponds to one of the wavelengths at which emission takes place. If a low-density gas is silhouetted against a source of a continuous spectrum, it absorbs light at certain wavelengths to produce a series of dark **absorption lines**. A typical star has an **absorption-line spectrum** (a continuous spectrum with dark lines superimposed by its atmosphere), whereas an emission nebula has an **emission-line spectrum**. See also *spectral line*.